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Electro-optically spectrum switchable, multiwavelength optical parametric oscillators based on aperiodically poled lithium niobate

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We report the first fast switchable multiwavelength optical parametric oscillator based on aperiodic optical superlattice technology. The constructed aperiodically poled lithium niobate (APPLN) integrates the functionalities of two quasi-phase-matching devices on a chip to work simultaneously as an electro-optic (EO) switchable notch-like filter and a multiline optical parametric downconverter. When such an APPLN is built in a 1064-nm-pumped optical resonator system, we achieve the oscillation of dual signals at 1540 and 1550 nm, for a single signal at 1540 nm, and a single signal at 1550 nm in the system when the 3-cm-long APPLN is driven by 0 V, 354 V, and 805 V, respectively. The switching among the three signal spectra is operationally simple and electro-optically fast. The electro-optically switched signals also feature enhanced power spectral density due to the unique EO gain-spectrum filtering mechanism employed in this work. © 2020 Optical Society of America

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Multiwavelength coherence light sources have been demonstrated in laser and optical parametric oscillator (OPO) systems [1–4]. In contrast to lasers, OPO technology features important advantages of providing a continuously and broadly accessible spectral range for the new wave generations. Besides, the addressable spectral region in an OPO can be shifted with the pump wavelength to cover a desired band according to the energy conservation law. The generation of a specific spectrum (including a multiline spectrum) in an OPO can be possibly achieved in the broad downconversion spectral region or within the broad parametric gain bandwidth simply via the preparation of appropriate phase-matching conditions (and the corresponding nonlinear gains) in a nonlinear-optic material, especially utilizing quasi-phase-matching (QPM). With the advances of the QPM technology, the featured high flexibility of domain engineering in materials such as lithium

niobate (LiNbO₃) enables the creation of a desired phase-matching condition/spectrum for specific wave mixings [5,6]. Multiwavelength OPOs have been demonstrated based on cascade-grating periodically poled lithium niobate (PPLN) and aperiodically poled lithium niobate (APPLN) crystals [3,4,7]. An APPLN, constructed using the aperiodic optical superlattice (AOS) technology [8], becomes a more favorable solution for realizing multiline sources using nonlinear frequency mixing due to its great ability to be tailored for a desired conversion spectrum and relatively high overall conversion efficiency [9].

Multiline OPOs are attractive for many applications including optical communications, remote sensing, LIDAR, biomedical measurement, and terahertz wave generation. The ability to switch wavelengths of those multiline systems is highly desirable for increasing their multiplexing, functionality, and applicability. Unfortunately, switchable multiline sources are not widely demonstrated except those based on fiber lasers [10]. The wavelength switching mechanisms developed in those fiber-optic systems are usually not convenient to be implemented in most existing OPOs. Optical elements such as etalons, gratings, or filters could be alternatives for OPO signal tunings but are based on unfavorably slow mechanical modulations. Electro-optic (EO) mechanisms have been used to build fast modulators with a bandwidth of > tens of GHz [11]. The EO effects of PPLN/APPLN have been applied to realize OPOs with fast spectral tuning or tailored gain-spectra [4,5]. However, a multiline source whose spectrum can be switched (electro-optically) fast has not yet been demonstrated.

In this study, to achieve the spectrum switching in a QPM multiline OPO, we directly build an electro-optically controllable parametric-gain suppression mechanism in the QPM medium to work like a notch filter to filter out those unwanted signal lines to prevent them from being built up in the OPO. This mechanism accesses the off-diagonal EO coefficient r_{51} of LiNbO₃ whose sign is spatially modulated, enabling the energy exchange between the extraordinarily (e) and ordinarily (o) polarized modes of waves at specific wavelengths satisfying the

QPM conditions created in the modulated superlattice. The conversion efficiency of an e -polarized mode to its o -polarized mode (and vice versa) in such a polarization mode converter (PMC) can be described by the expression derived based on a PPLN scheme [12]:

$$\eta_{\text{pmc}} = |\kappa|^2 L^2 \text{sinc}^2(\sqrt{|\kappa|^2 + (\Delta\beta/2)^2} L) = 1 - T_{\text{pmc}}, \quad (1)$$

which describes the power exchange rate of the device (T_{pmc} denotes the power ratio of the wave remaining at the e -polarization state), where L is the device length, $\Delta\beta$ is the wave-vector mismatch of the conversion process, and $|\kappa| = (\pi/\lambda_0) n_{o,\text{eff}}^{3/2} n_{e,\text{eff}}^{3/2} |r_{51} E_y G_m|$ is the coupling coefficient. This is a function of the wavelength, λ_0 , the effective refractive indices of the o and e polarized modes, $n_{o,\text{eff}}$ and $n_{e,\text{eff}}$, the EO coefficient, r_{51} , the external electric field along the crystallographic y axis, E_y , and the reduction factor (Fourier coefficient) associated with the Fourier representation of the m th-order QPM grating structure, G_m . The power exchange between the two polarization modes varies periodically (in a sine-squared function) with E_y according to Eq. (1) when the QPM condition is satisfied (i.e., $\Delta\beta = 0$), as the solid blue curve shown in Fig. 1 for a 1550-nm wave (e -polarized initially) after traversing a 3-cm-long PPLN of a 50% domain duty cycle. Interestingly, this electric-field tuning curve is a function of G_m which is determined by the domain structure of the QPM device according to Eq. (1). For example, the dashed red curve in Fig. 1 shows the tuning curve when the PPLN device is now structured to be with a 23%/77% duty cycle. This implies a specific power exchange state at a specific E_y can be possibly attained with such a QPM PMC via the use of a proper domain structure.

In a type-0 (eee) QPM optical parametric downconverter (OPDC) that accesses the largest nonlinearity (d_{33}) of LiNbO₃, the generated signal (and idler) waves are in e -polarized modes. If the PMC mechanism is built in an OPO based on the type-0 OPDC, those generated e -polarized signals within the conversion bandwidth (2 nm-cm at a 1.5- μm band [12]) of the PMC can thus be (completely or significantly) converted to o polarization modes and then fail to build up in the OPO owing to the resulting large phase mismatch. While the devices studied in [4,5] behave like a fine notch filter to filter out certain spectral content of the OPO signals, in this study, the challenge is to develop a device capable of working like an active notch filter whose spectral filtering position can be fast EO-switched among some specific wavelengths (i.e., OPO signal lines). As that shown in Fig. 1, the electric-field tuning curve of a PMC device depends on the domain-structure factor G_m

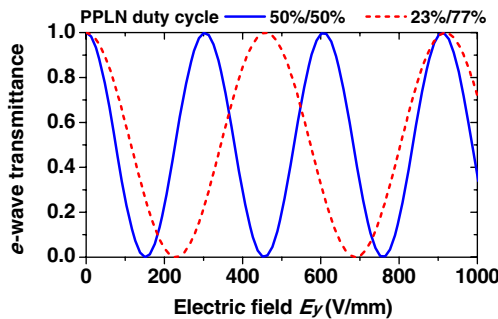


Fig. 1. Calculated power transmittance of an e -polarized 1550-nm mode in PPLN PMCs of different domain duty cycles versus E_y .

which is wavelength-dependent as well. Two signal waves of different wavelengths but with the same polarization modes can turn out to be in different modes (i.e., in e - and o -polarized modes, respectively) after interacting with a properly domain-structured device under a specific E_y and thus one (e -polarized mode) will experience gain and be amplified in the OPO while the other (o -polarized mode) will be highly phase-mismatched and suppressed. Switching among different OPO signals can be implemented when their respective electric-field tuning curves can be appropriately created.

In this work, the functionality of an EO PMC-based spectrum switchable notch filter is to be integrated with that of a multiwavelength OPDC in an optical superlattice LiNbO₃ substrate. We have developed an algorithm based on the coupled-mode theory [5] and the AOS technique with the aid of the simulated annealing (SA) optimization method [13]. In the calculation, a z -cut LiNbO₃ crystal consisting of a number of N domain blocks is assumed. Each block has a thickness of Δx and a domain polarity of either positive or negative corresponding to the $+z$ or $-z$ crystal orientation, respectively, which is to be determined by the developed algorithm. In this study, the target of the calculation is to find an APPLN domain structure to perform simultaneously the two device functionalities in a 1064-nm-laser-pumped OPO to realize the generation and switching of dual signal lines at 1540 and 1550 nm. To help best achieve the target, the following objective function (OF) was implemented in the algorithm:

$$\begin{aligned} \text{OF} = & \left(\sum_{i=1}^M w_{\text{op}}(\lambda_{s,i}) \left| \eta_{\text{op},0}(\lambda_{s,i}) - \eta_{\text{op},c}(\lambda_{s,i}) \right| \right. \\ & \left. + w_{\text{pmc}}(\lambda_{s,i}) \left| T_{\text{pmc},0}(\lambda_{s,i}, E_r) - T_{\text{pmc},c}(\lambda_{s,i}, E_r) \right| \right) \\ & + \gamma_{\text{op}} (\max[\eta_{\text{op},c}(\lambda_{s,1}), \dots, \eta_{\text{op},c}(\lambda_{s,M})] \\ & - \min[\eta_{\text{op},c}(\lambda_{s,1}), \dots, \eta_{\text{op},c}(\lambda_{s,M})]), \end{aligned} \quad (2)$$

where M is the number of the working bands for the OPDC and EO PMC integrated device, $\lambda_{s,i}$ is the wavelength of the i th working band, and $\eta_{\text{op},c}(\lambda_{s,i})$ and $T_{\text{pmc},c}(\lambda_{s,i})$ are the calculated conversion efficiencies of the OPDC and EO PMC (at $E_y = E_r$) at $\lambda_{s,i}$, respectively, while $\eta_{\text{op},0}(\lambda_{s,i})$ and $T_{\text{pmc},0}(\lambda_{s,i})$ are their respective target values, $w_{\text{op}}(\lambda_{s,i})$ and $w_{\text{pmc}}(\lambda_{s,i})$ are the weighting factors for the efficiency of the i th conversion process of the OPDC and EO PMC, respectively, γ_{op} is the weighting factor for a peak-efficiency-equalized multiple OPDC process, and the operators $\max[\dots]$ and $\min[\dots]$ select the maximum and minimum from the data included in the square bracket. In this work, $M = 2$ is used for a switchable dual-signal OPO with wavelengths at $\lambda_{s,1} = 1540$ nm and $\lambda_{s,2} = 1550$ nm.

Another aim of this calculation is to find the two lowest possible operating voltages, V_1 and V_2 ($V = E_y \times d$ with $d = 1$ mm being the electrode spacing), for the APPLN chip to switch the OPO among three possible emission spectra, i.e., 0 V (V_0) for dual wavelengths (1540 and 1550 nm), V_1 for a single wavelength at 1540 nm, and V_2 for a single wavelength at 1550 nm. This is to find appropriate values for $T_{\text{pmc},0}(1540 \text{ nm})$

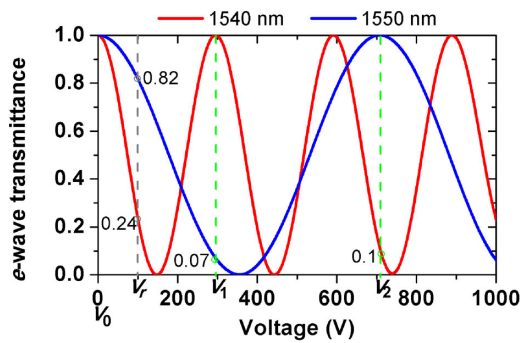


Fig. 2. Calculated round-trip electric-field tuning curves of a 3-cm-long APPLN for $\lambda_{s,1} = 1540$ nm and $\lambda_{s,2} = 1550$ nm. V_1 and V_2 indicate the optimal EO switching voltages.

and $T_{\text{pmc},0}(1550 \text{ nm})$ at an arbitrary reference driving field $E_r = V_r/d$ to help create electric-field tuning curves for the 1540- and 1550-nm waves with a desired relationship. Figure 2 shows the calculated electric-field tuning curves of a 3-cm-long APPLN chip for the two signal waves (*e*-polarized initially). The domain structure of the chip is calculated from the developed SA algorithm that considers waves making only one cavity transit (i.e., one round trip; $p = 1$) in the OPO to expedite the optimization process. The optimal domain structure of the chip (calculated based on $p = 1$) has been further confirmed to perform the target spectral switching in a multitransit OPO by simulation. The curves in Fig. 2 clearly show the two signals are both *e*-polarized at 0 V (V_0) before any EO control, and both can thus experience gain and be amplified in the OPO. One can further find two particularly interesting operating conditions from the curves, where the device will output a 100% *e*-polarized ($T_{\text{pmc}} = 1$) 1540-nm signal and a 93% *o*-polarized ($T_{\text{pmc}} = 0.07$) 1550-nm signal at $V_1 = 295$ V, while it will yield a 100% *e*-polarized 1550-nm signal and a 90% *o*-polarized ($T_{\text{pmc}} = 0.1$) 1540-nm signal at $V_2 = 708$ V, as indicated by the two dashed green lines in Fig. 2. Another dashed line indicated at $V_r = 100$ V reveals $T_{\text{pmc},e}(1540 \text{ nm}) \sim 0.24$ and $T_{\text{pmc},e}(1550 \text{ nm}) \sim 0.82$ which are the values set in the algorithm [see Eq. (2)] to define the respective periodicities of the two sine-squared tuning curves to reach the targeted voltage-dependent polarization-mode switching relationship between the two signal waves. The result in Fig. 2 implies an OPO whose output is switchable among three different spectra: dual 1540- and 1550-nm lines, a single 1540-nm line, and a single 1550-nm line can be possibly achieved in this APPLN device when operated at V_0 , V_1 , and V_2 , respectively, as targeted.

Figure 3 shows the Fourier spectrum of the calculated APPLN domain structure. Four prominent spectral peaks at spatial frequencies of 0.03330, 0.03342, 0.04713, and 0.04747 μm^{-1} are clearly resolved. The first two peaks are associated with the reciprocal vectors formed by the domain structure for quasi-phase-matching the 1064-nm-pumped 1550/3393 and 1540/3442-nm signal/idler OPDC processes at 34°C, while the latter two peaks emerge as expected to phase-match the corresponding EO PMC processes for performing the targeted signal-spectral switching. In contrast to the even-height Fourier peaks obtained for reaching an equal efficiency for the two OPDC processes, the heights of the two EO PMC peaks are in a certain ratio (of G_m) for achieving the desired tuning curves with proper periodicities as discussed earlier. The inset (i1)

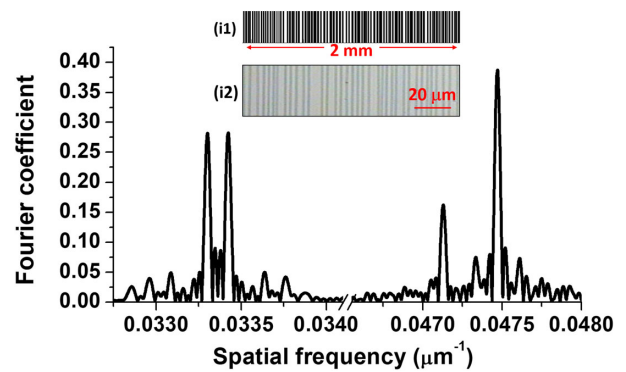


Fig. 3. Fourier spectrum of the APPLN domain structure. Four prominent spectral peaks, required for phase-matching the dual-wavelength OPDC and EO PMC processes, are resolved. The insets show a section of the (i1) calculated and (i2) fabricated domains.

shows the calculated domain polarity distribution of a section of the APPLN, where the finest domain width is $\Delta x = 5 \mu\text{m}$.

We fabricated the APPLN in a 30-mm-long, 1-mm-wide, and 0.5-mm-thick *z*-cut LiNbO₃ chip by using the standard electric-field poling technique. The engineered domains exhibit a randomly underpoled error [14] of a maximum of 45% over the poled area. A typical image of the poled domains taken from a portion of the hydrofluoric acid (HF)-etched +*z* surface of the fabricated APPLN is as shown in the inset (i2) of Fig. 3. The end (*x*) faces of the APPLN chip were polished and applied with antireflection coatings for wavelengths at 1064, 1540, and 1550 nm. The two *y* faces of the chip were sputtered with a NiCr alloy to form the electrodes for the E_y application. The APPLN, temperature controlled at 34°C, is then placed in a 53-mm-long linear resonator formed by a 100-mm radius-of-curvature meniscus BK7 mirror (M1) and a plane-plane output-coupling mirror (M2). The pump laser for this OPO is a linearly polarized, actively *Q*-switched 1064-nm Nd:YAG laser (13.6-ns pulse width and 1-kHz repetition rate, $M^2 < 2$). M1 has $\sim 99\%$ reflectance over the signal range 1520 to 1560 nm and $\sim 92\%$ transmittance at 1064 nm, while M2 is $\sim 94\%$ reflective at 1540 and 1550 nm and $\sim 96\%$ transmissive at 1064 nm. A convex lens of 250-mm focal length is used to couple the pump laser into the OPO. Figure 4 shows the measured (solid lines) signal spectra of the OPO under applied voltages of 0, 354, and 805 V, respectively, at a pump peak power of 9.6 kW with an optical spectrum analyzer (Anritsu MS9740A) of 0.07-nm resolution. It clearly shows the OPO output can be electro-optically switched among three different spectra of dual 1540 and 1550 nm, single 1540 nm, and single 1550-nm signal lines. The demanded switching voltages are increased due to the domain poling error found in the APPLN. The error is also responsible for the uneven spectral heights measured with the dual-wavelength signal at 0 V [Fig. 4(a)]. We also observed a weak non-phase-matched sum frequency generation (SFG) [together with the second harmonic generations (SHG)] of the dual signal lines generated through the APPLN [see the inset of Fig. 4(a)], manifesting well the synchronization of the two emission lines. The dashed lines in Fig. 4 are results from the calculation model developed in [5]. We obtain the best fits for these experimental results when the APPLN device with an average domain error of 27% and a number of $p = 9, 10$, and 10 cavity transits for the 0-, 354-, and 805-V signal buildup,

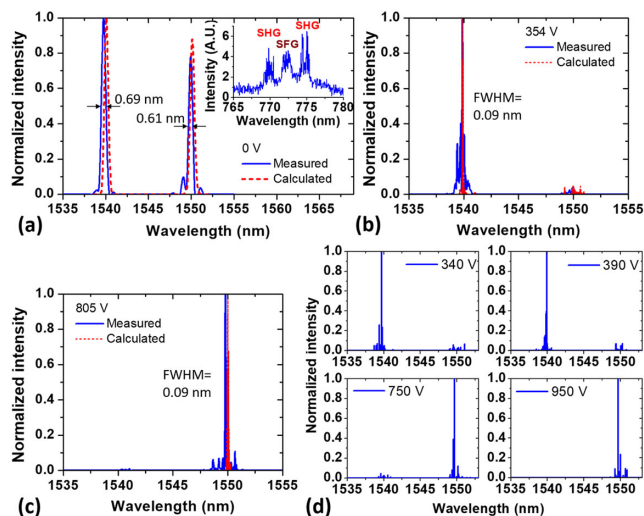


Fig. 4. Measured and calculated signal spectra of the OPO under applied voltages of (a) 0, (b) 354, and (c) 805 V at 9.6-kW pump power. (d) Measured EO switched spectra at various voltages. The inset in (a) shows the measured SFG and SHG of the dual lines.

respectively, is used. There exists a certain discrepancy between the simulated and measured results, which can be attributable to our calculation model being based on the plane-wave approximation, while the experimental OPO system involves the Gaussian beam and even some higher-order spatial modes. A similar issue on an EO PPLN device has been studied [15].

In Figs. 4(b) and 4(c), one signal is successfully suppressed as expected, while the other one emerges with an increased peak spectral intensity as a result of the combination of the narrow bandpass filtering characteristic of the EO PMC and the nonlinear parametric gain effect [4,5]. According to Figs. 4(b) and 4(c), the extinction ratios of the 1550- and 1540-nm signals (in contrast to their unsuppressed counterpart) are estimated to be about -15 and -21 dB, respectively. A more complete extinction of the EO suppressed signals can be possibly achieved by creating broader bandwidths for those spectral peaks associated with EO PMC processes. The measured output pulse energies and pulse widths (over the whole signal spectrum) for the three spectra obtained at 0, 354, and 805 V are ~ 1.07 , 0.81, and 0.90 μJ and ~ 7.0 , 7.5, and 7.4 ns, respectively. Though the peak powers of the three switchable outputs are similar, the power spectral densities of the signals obtained by EO switching are remarkably increased by >5 times compared to that obtained at 0 V.

Because of the domain poling error, the tuning curves in Fig. 2 are shifted, resulting in $T_{\text{pmc}} = \sim 0.99$ and ~ 0.26 at

354 V and $T_{\text{pmc}} = \sim 0.24$ and ~ 1 at 805 V for 1540- and 1550-nm waves, respectively. This implies the OPO can work over a range of fabrication error of the core device. Figure 4(d) shows the single-line spectra of the OPO over broad voltage ranges.

In conclusion, we have demonstrated a multiline OPO whose spectrum is electro-optically switchable, achieved using an integrated APPLN photonic chip. The APPLN, constructed using a SA algorithm, is the core device to create appropriate EO parametric gain tuning curves for the selection of one or more signal waves in the OPO simply by applying appropriate voltages. When the APPLN is operated at 354 V and at 805 V, the OPO signal spectrum can be switched from originally dual 1540- and 1550-nm lines at 0 V to a single 1540-nm line and a single 1550-nm line, respectively, with a >5 -fold enhancement of the power spectral density.

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Disclosures. The authors declare no conflicts of interest.

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